Fatemeh Shafizadeh Ghahderijani

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Education

University of Cologne — Cologne Germany
M.Sc. in Physics — *Oct 2025 – Presnet*

Shahid Beheshti University — Tehran, Iran
B.Sc. in Physics; GPA: 17.71/20 (German equivalent: 1.8) — *Sep 2020 – Sep 2025*

Research Interests

- Theoretical Condensed Matter Physics
- Quantum Foundation
- Mathematical Physics
- Quantum Information and Computation

Research Experience

Research Assistant — Tehran, Iran
Shahid Beheshti University — *Oct 2023 – Mar 2025*

- Implemented quantum machine learning models through the development of Quantum Restricted Boltzmann Machines (QRBMs) under the supervision of *Dr. S.M.H. Halataei*.
- Focused on enhancing computational accuracy and time performance, achieving significant gains in processing efficiency and model reliability compared to standard classical frameworks.

Work Experience

Teaching Assistant — Tehran, Iran
Shahid Beheshti University — *Oct 2023 – Jun 2025*

- Solid state Physics I, Assistant to *Dr. M. Hafez-Torbati* (Summer Semester 2024)
- Physics I, Assistant to *Dr. M. Hafez-Torbati* (Winter Semester 2023, 2024 and Summer Semester 2025)

Projects

Classification of Pottery Motifs Using Machine Learning

- Collaborated with a multidisciplinary team to integrate computational methods into archaeological research.
- Applied machine learning techniques to classify pottery motifs and distinguish between three cultural origins.

Relevant Courses and Grades

Quantum Computing *(Graduate-Level Course)*	18/20
Quantum Mechanics I	19.5/20
Quantum Mechanics II	18.5/20
Solid State Physics I	17.25/20
Physics Project	20/20
Group Theory	*18.4*
Foundation of Numerical Simulations	*18.5*
Mathematical Physics I	16.3/20
Mathematical Physics II	20/20
Mathematical Physics III	17.7/20
Thermodynamics and Statistical Mechanics I	19.2/20
Thermodynamics and Statistical Mechanics II	18.75/20
Biophysics	19.4/20

For the full grades, see here.

Volunteer Experience

Member of Scientific Association of Physics students Tehran, Iran
Shahid Beheshti University *Nov 2022 – Sep 2023*

- Collaborated with the executive team to plan and organize events, including workshops and social gatherings.
- Contributed innovative ideas and strategic insights during weekly meetings to improve the association's impact and operational efficiency.

Representative of the Department of Physics to the University President Tehran, Iran
Shahid Beheshti University *Nov 2023 – Present*

- Acted as a student representative, voicing student interests and concerns within the Faculty of Physics.
- Improved the academic environment through collaboration with faculty and administration, advocating for enhanced educational resources and support for students.

Content writer – Quark Scientific journal Tehran, Iran
Shahid Beheshti University *Nov 2023 – May 2024*

- In the second issue of the journal, I translated a report about New Type of Entanglement that allows scientists to see inside nuclei.
- In the third issue of the Journal, our group wrote about Quantum Teleportation.

Awards

Top-Ranked Student Award: Ranked among the top 10% of students in the Department of Physics for 2023 and 2024, based on GPA.

Skills

Programming: Python, Mathematica

Tools and Technologies: NumPy, Matplotlib, Pandas, PyTorch, Git, LaTeX, Microsoft Office

Soft Skills: Time Management, Teamwork, Problem-Solving, Literature Review

Additional Skills: Playing the piano

Languages: Persian (Native), English (TOEFL iBT 95)

Talks

Modern Machine Learning and Particle Physics Tehran, Iran
Shahid Beheshti University, Faculty of Physics *Dec 2024*

Restricted Boltzmann Machine Derivations Tehran, Iran
Sharif University of Technologies, Center of Quantum Sciences *Sep 2024*

Workshops and Conferences

ML4Q Platforms for Quantum Technologies Course *Feb 2026*
Organized by ML4Q Cluster of Excellence , NRW, Germany

Quantum Photonic Technologies Winter School *Mar 2024*
Organized by Laser and Plasma Research Institute, Shahid Beheshti University, Tehran, Iran

Second Quantum Technologies Workshop *Mar 2024*
Organized by PsiKet Academey, Sharif University of Technologies, Tehran, Iran

Quantum Computing based on Superconducting Circuits Conference *Feb 2024*
Organized by IQTEC and the Faculty of Physics, Kharazmi University, Tehran, Iran

Cooperation with SESAME International Laboratory Conference *Feb 2024*
Organized by Institute For Research In Fundamental Sciences (IPM), Tehran, Iran

The 28th Special School on Topics in Physics *July 2023*
Organized by Institute for Advanced Studies in Basic Sciences (IASBS) in Zanjan, Iran